TECHNICAL ASSESMENT

PYQs & Key Concepts

What will be the output of the following pseudocode ?

```
Set Character m='a'
Set Integer count=0
while(m>95)
   count=count+1
   m=m-1
end-while
display count
```

a) 0

b) 1

c) 2

d) 3

What will be the output of the following pseudocode ?

```
Set Integer array a=[1,2,3,4,5]
Set Integer array b
Set Integer len=ArrayLength(a)
for c = len-1 to 0 step -1 and d = 0 to len-1 step 1 do
  b[d] = a[c]
end-for
display b
```

a) [4, 5, 1, 2, 3]

b) [3, 4, 5, 2, 1]

c) [5, 4, 3, 2, 1]

d) [1, 5, 2, 3, 4]

What will be the output of the following pseudocode ?

```
Set x = 'a' * 10
print x
```

a) 97

b) 970

c) 'a' repeated 10 times

d) Error

What will be the output of the following pseudocode for n=35?

```
Read n
i = 0
While n % 10 != 0
n = n + 3
i++
end while
n = n+i
print n
```

a) 55.0

b) 45.0

c) 50.0

d) 43.0

What will be the output of the following pseudocode ?

```
SET ARR = [2, 4, 6, 8, 10]
SET SUM = 0
FOR I = 1 TO LENGTH(ARR)
    IF ARR[I] % 4 == 0 THEN
        SUM = SUM + ARR[I]
    END IF
END FOR
PRINT SUM
```

A) 14

B) 12

C) 18

D) 20

What will be the time complexity of the following code?

```
value = 0
for i in range(n):
    for j in range(i):
        value = value + 1
```

A) **n**

B) **(n + 1)**

C) n(n - 1)/2

D) n(n + 1)

What will be the output of the following pseudocode for n=10?

```
Read n
Initialize i to 5, sum to 0
while i < n do
   increase sum by i
   increment i
end while
Write sum
```

a) 35.0

b) 25.0

c) 45.0

d) 55.0

What will be the output of the following pseudocode ?

```
SET N = 4
SET ARR = [1, 2, 3, 4]
SET SUM = 0
FOR I = 1 TO N
    FOR J = I TO N
        SUM = SUM + ARR[J]
    END FOR
END FOR
PRINT SUM
```

A) 20

B) 25

C) 30

D) 35

What is the time complexity of the following code?

```
a = 0
for i in range(N):
    for j in reversed(range(i, N)):
        a = a + i + j
```

A) O(N)

B) O(N * log(N))

C) O(N * √N)

D) O(N * N)